want to learn more about Istio and service meshes in general.

Pop quiz

Q.5 Which of the following are reasons to use an intermediate CA? Select all that apply.

a. To have longer certificate chains
b. To keep your operations teams busy
c. To use smaller key sizes, which are faster
d. So that the root CA key can be kept offline
e. To allow revocation in case the CA key is compromised

Q.6 True or false: a service mesh can automatically upgrade network requests to use TLS?

Answers are at the end of the chapter.

10.3.3 Locking down network connections

Enabling TLS in the cluster ensures that an attacker can't modify or eavesdrop on communications between APIs in your network. But they can still make their own connections to any service in any namespace in the cluster. For example, if they compromise an application running in a separate namespace, they can make direct connections to the H2 database running in the `natter-api` namespace. This might allow them to attempt to guess the connection password, or to scan services in the network for vulnerabilities to exploit. If they find a vulnerability, they can then compromise that service and find new attack possibilities. This process of moving from service to service inside your network after an initial compromise is known as *lateral movement* and is a common tactic.